

OEAS 705/805 Advanced Environmental Data Science

Fall 2026

Week 5

Dr. Jessie Turner, jturners@odu.edu

Class times: OCNPS 320, T/Th, 11:00 - 12:15

Office hours: After class T/Th,

And: OCNPS 411, M, 1:00 - 3:00



Schedule

Homework 3 Due Monday Oct 19th

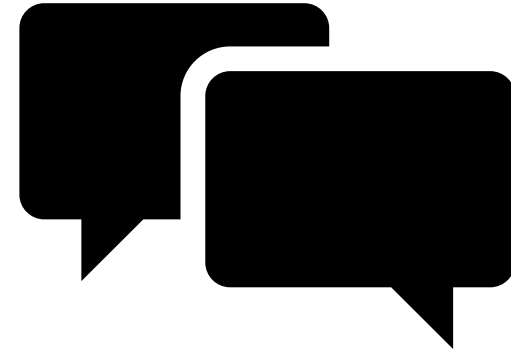
→ Post link to your OEAS805/HW3 GitHub to Canvas

Machine Learning – Supervised Learning until Oct 20

Machine Learning – Unsupervised Learning Oct 20, 22

Projects must be complete by X for Peer Reviews

Machine Learning



Discuss in pairs:

- What do you already know about machine learning?

Machine Learning



Machine Learning

Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and generalize to unseen data, and thus perform tasks without being explicitly programmed.

(Wikipedia)

Machine Learning

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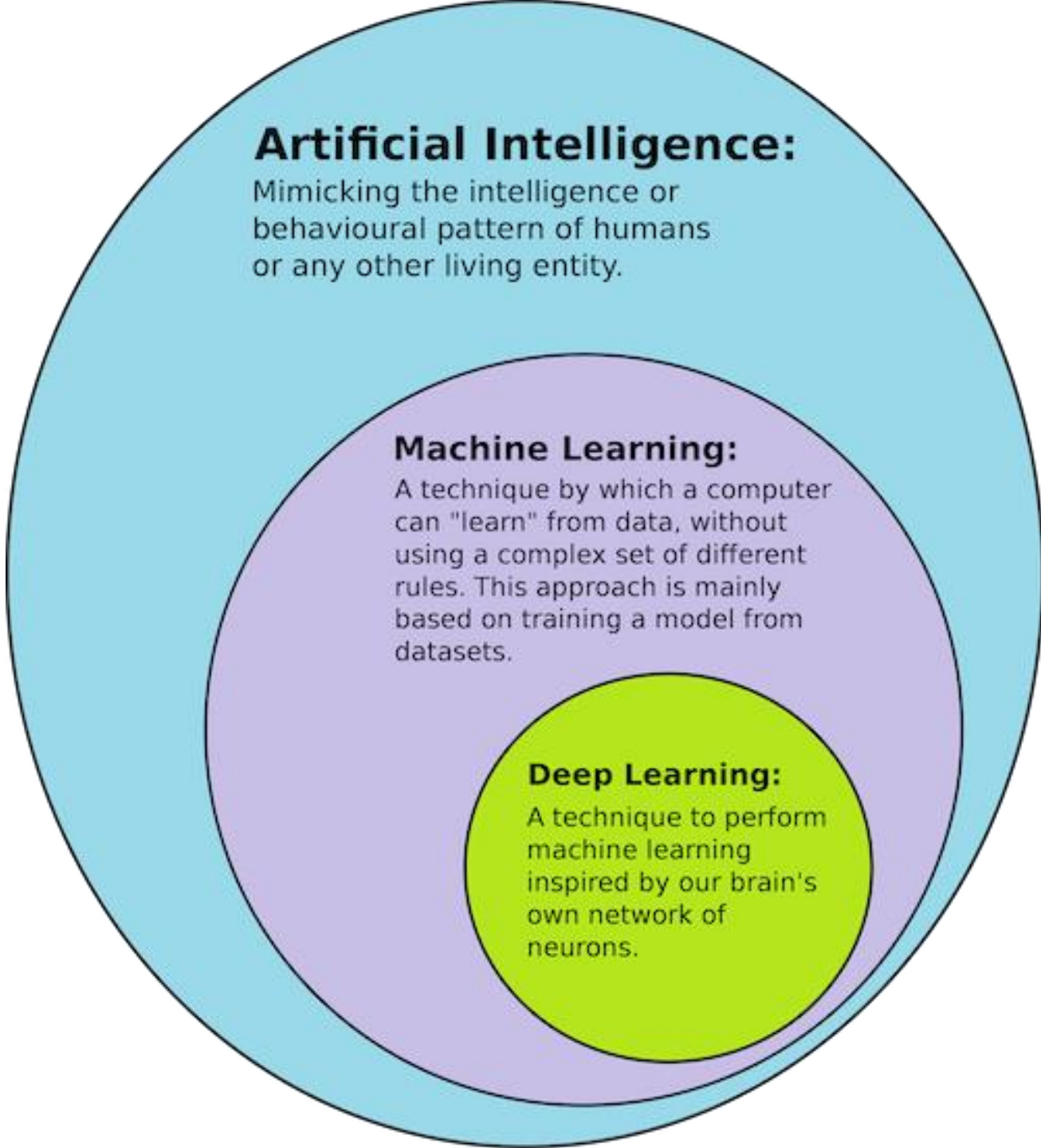
JUST ANOTHER STATISTICAL TECHNIQUE!

Machine Learning

Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and **generalize to unseen data**, and thus perform tasks without being explicitly programmed.

(Wikipedia)

OFTEN THE AIM IS TO MAKE PREDICTIONS



The diagram consists of three concentric circles. The outermost circle is light blue and contains the text for 'Artificial Intelligence'. Inside it is a medium-sized light purple circle containing the text for 'Machine Learning'. The innermost circle is light green and contains the text for 'Deep Learning'. This visualizes that Deep Learning is a subset of Machine Learning, which is a subset of Artificial Intelligence.

Artificial Intelligence:

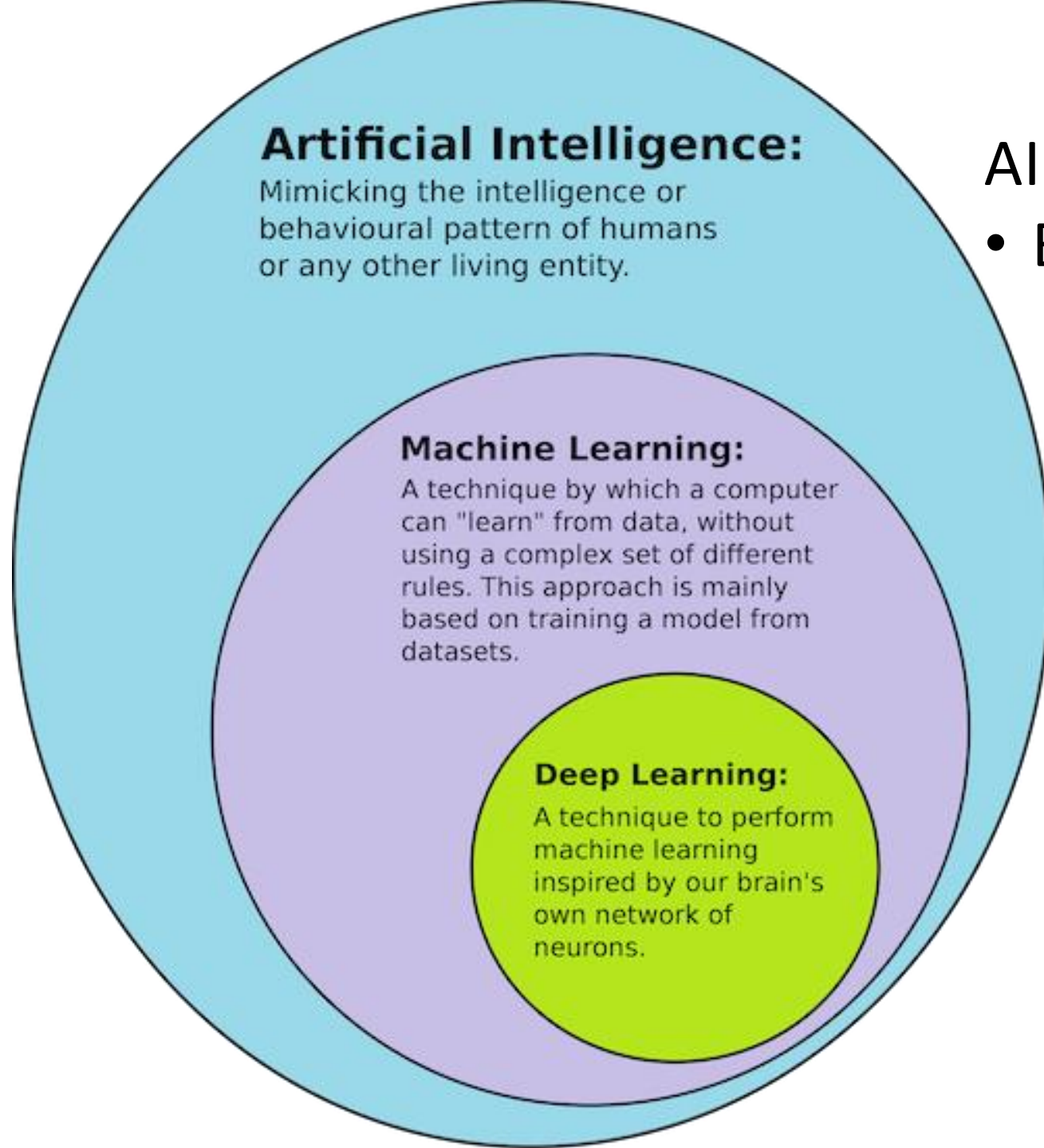
Mimicking the intelligence or behavioural pattern of humans or any other living entity.

Machine Learning:

A technique by which a computer can "learn" from data, without using a complex set of different rules. This approach is mainly based on training a model from datasets.

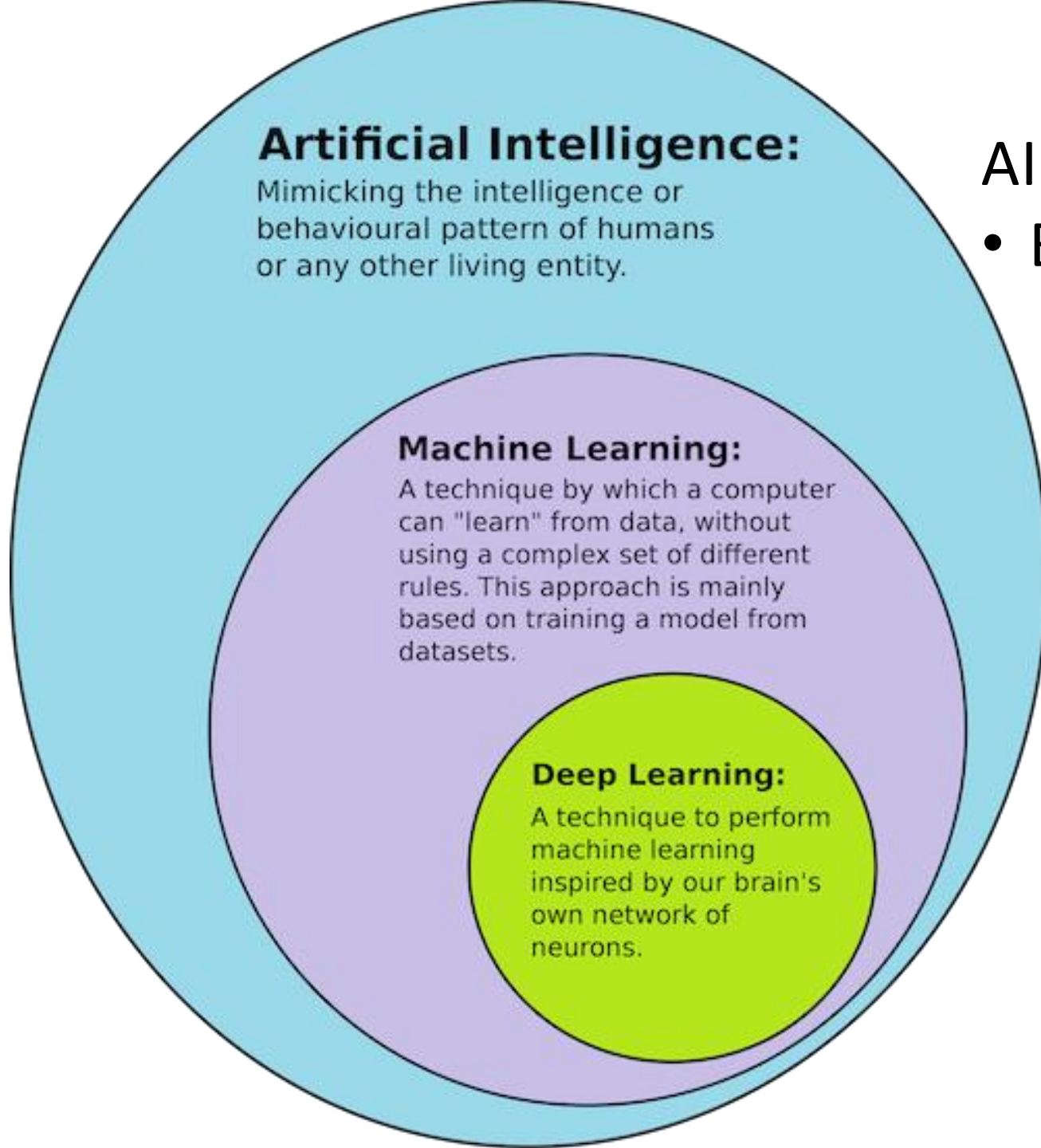
Deep Learning:

A technique to perform machine learning inspired by our brain's own network of neurons.



AI:

- Broad Term (Marketing)

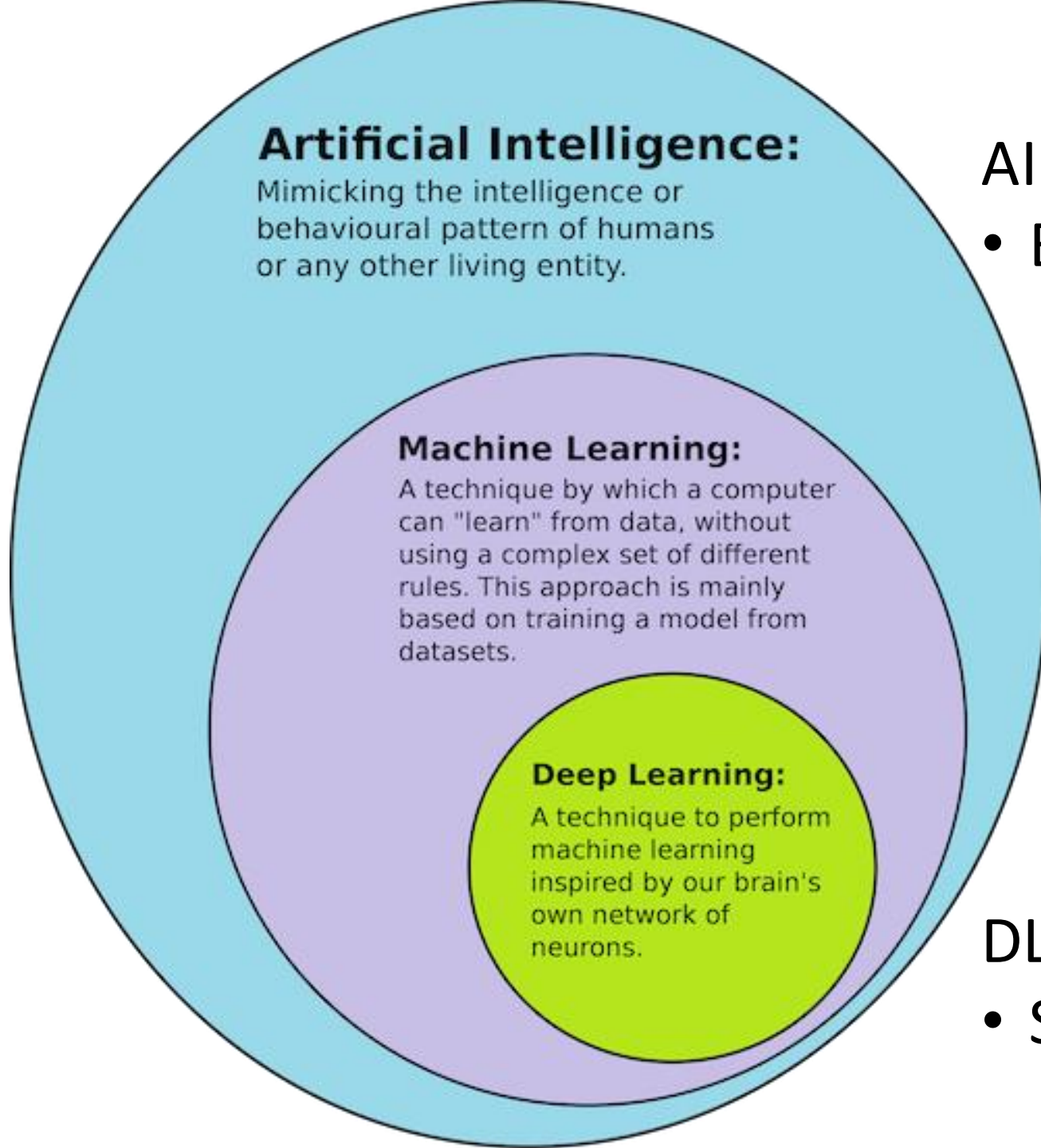


AI:

- Broad Term (Marketing)

ML:

- What we will be doing
- Multiple options in addition to neural networks



AI:

- Broad Term (Marketing)

ML:

- What we will be doing
- Multiple options in addition to neural networks

DL:

- Specifically neural networks

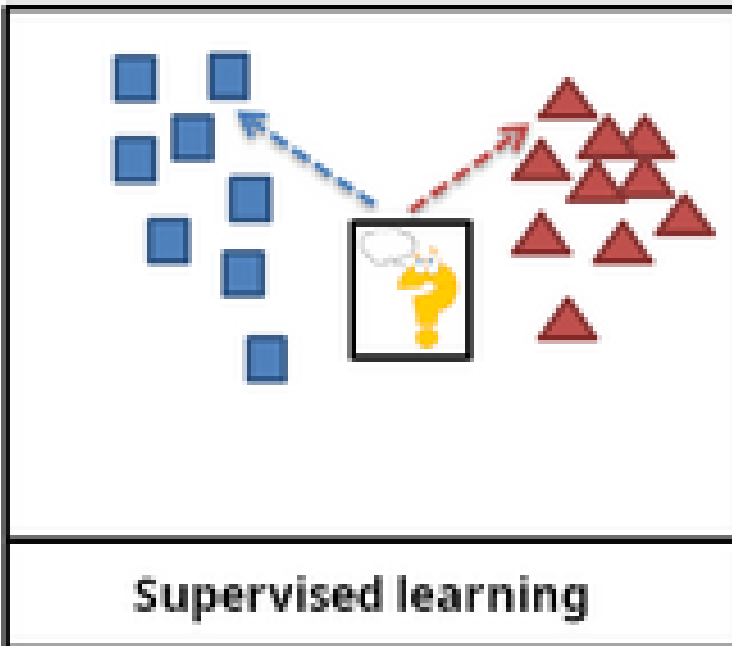
Machine Learning



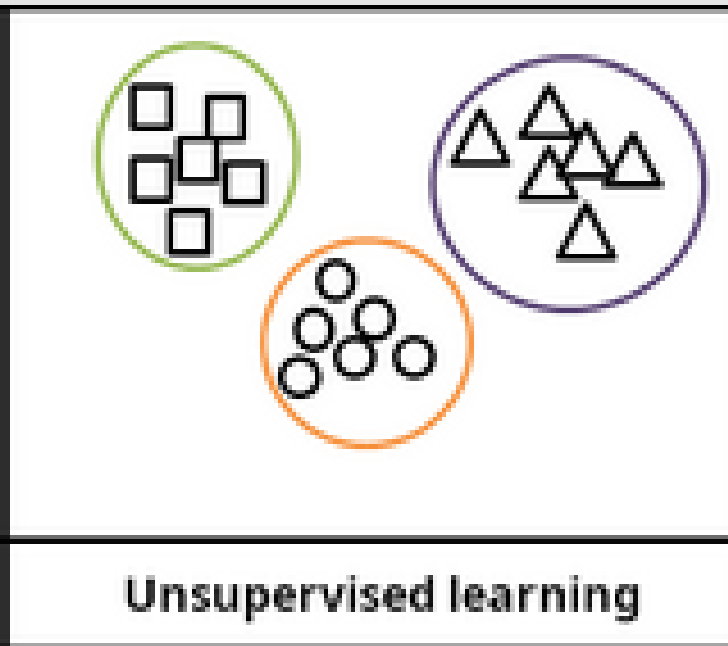
Gray et al. 2024, Oceanography: in “readings” folder on GitHub

Machine Learning

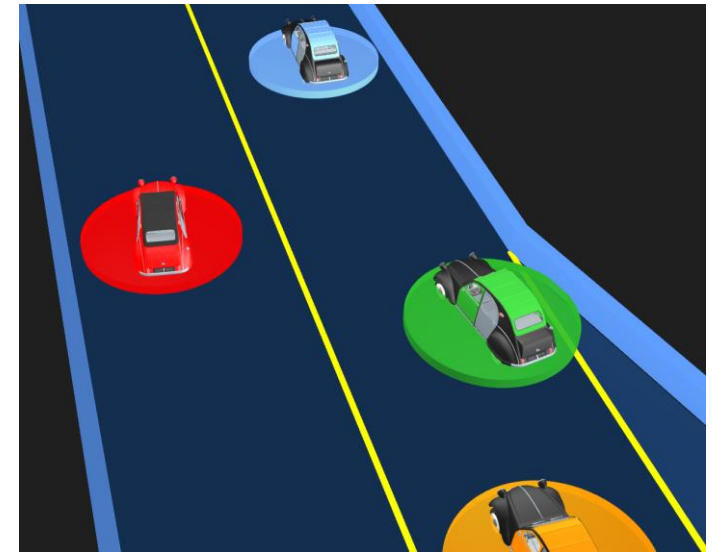
1. Supervised Learning



2. Unsupervised Learning



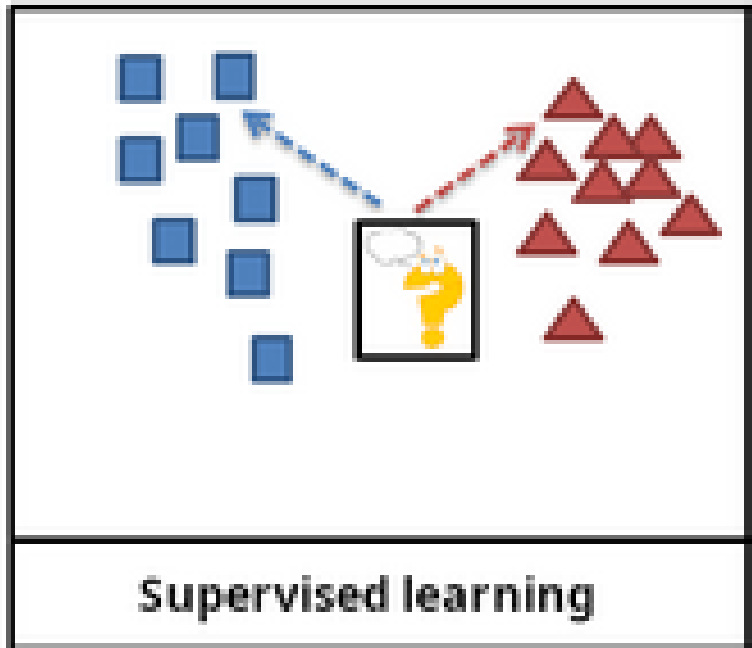
3. Reinforcement Learning



Game Theory applications,
Not as useful in earth sciences

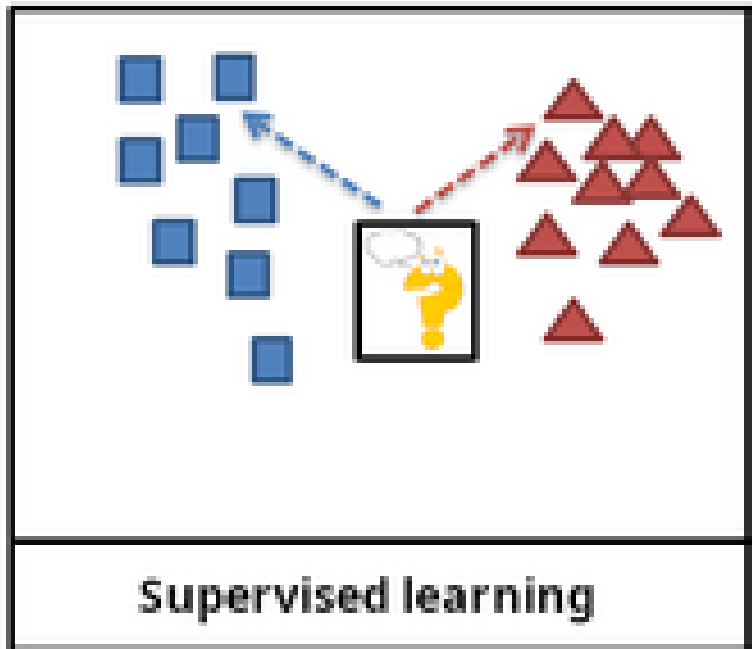
Machine Learning

1. Supervised Learning



Machine Learning

1. Supervised Learning

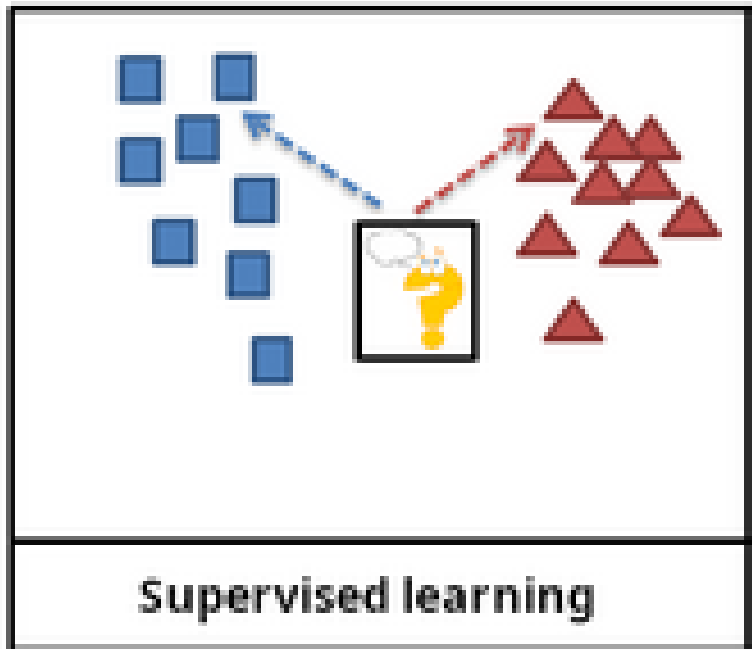


Labeled samples are used to train the model

- Regression
- Classification

Machine Learning

1. Supervised Learning

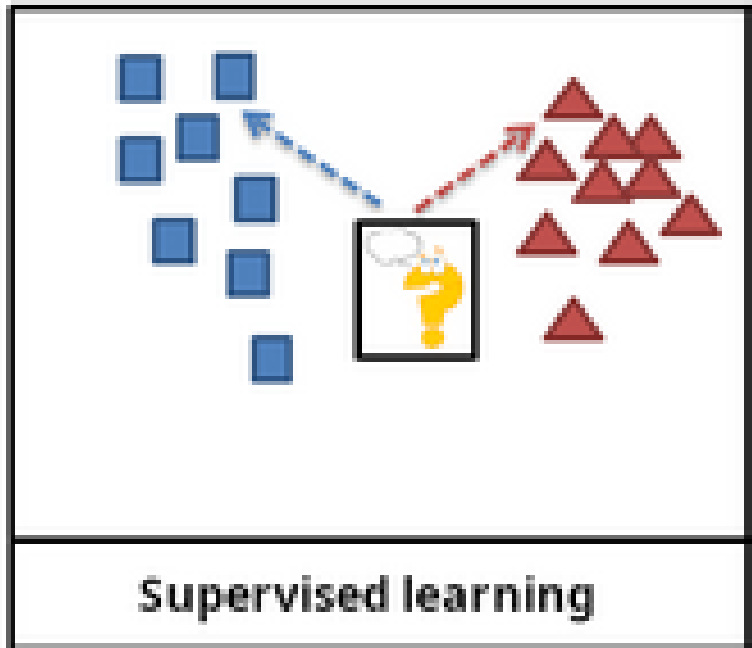


Labeled samples are used to train the model

- Regression – predict values
- Classification – predict category

Machine Learning

1. Supervised Learning



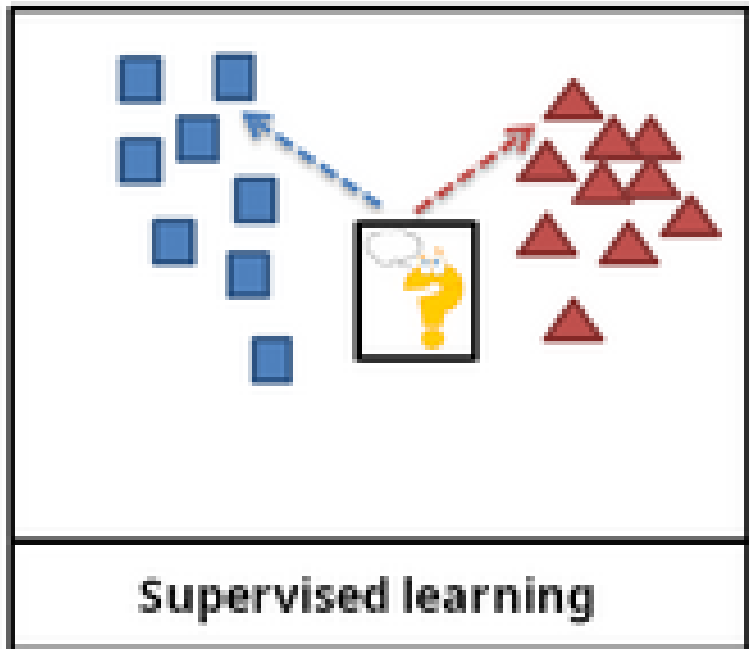
Examples:

- Random forests
- Support vector machines
- Gaussian processes
- Neural networks

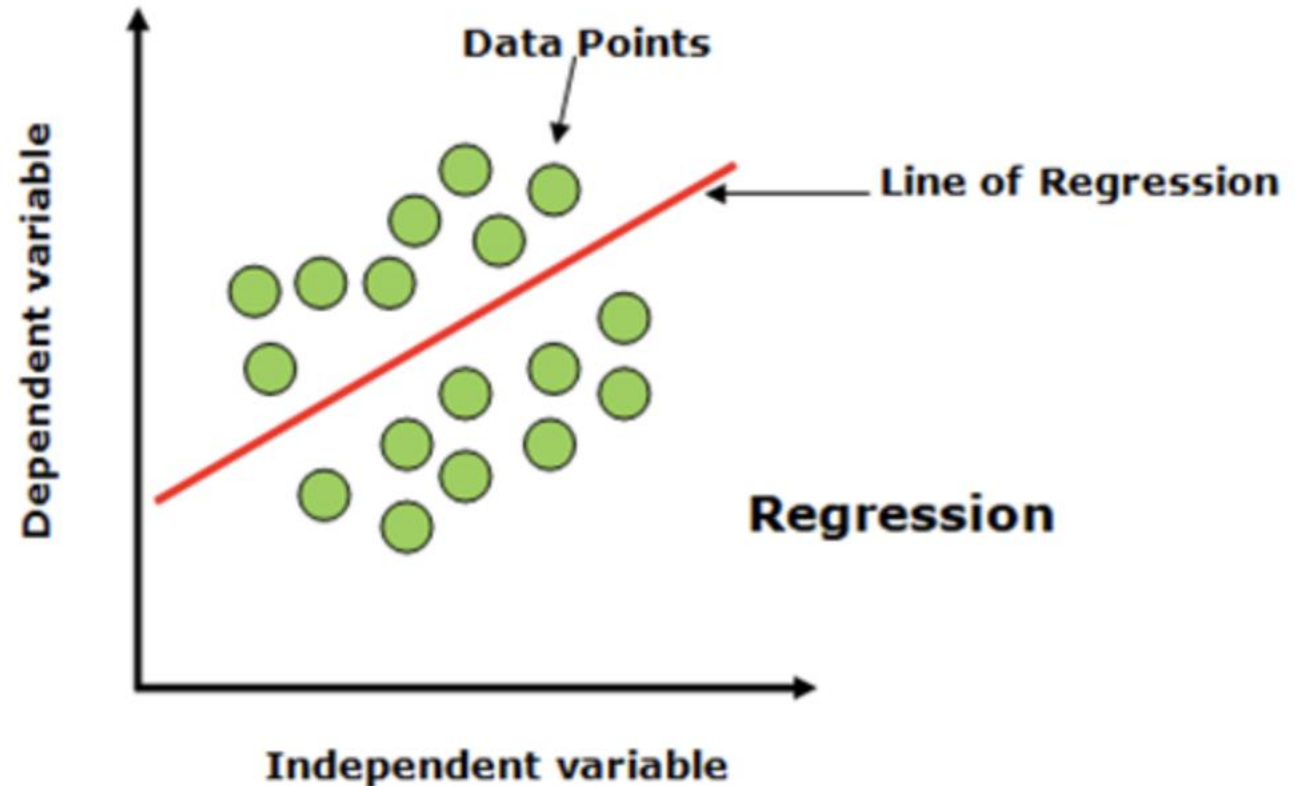
Machine Learning

You've done it before: linear regression

1. Supervised Learning

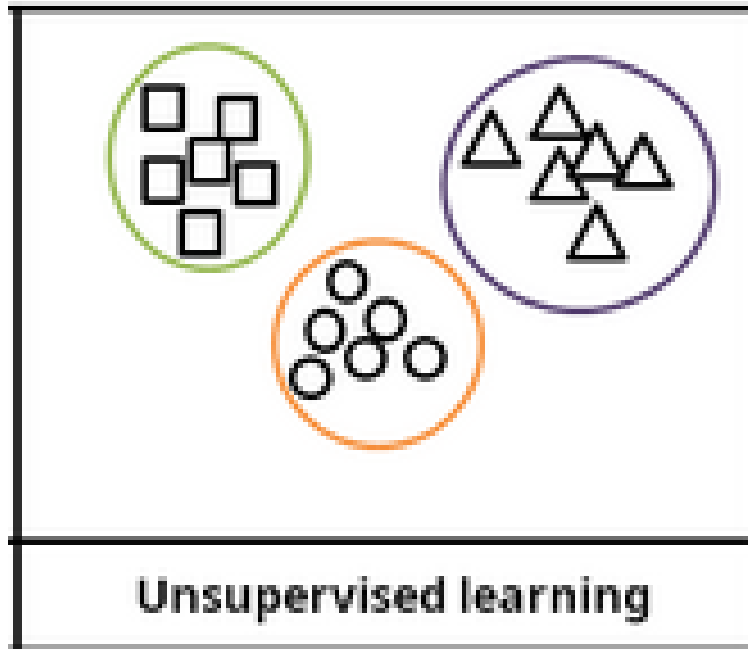


- One input (x)
- One output (y)



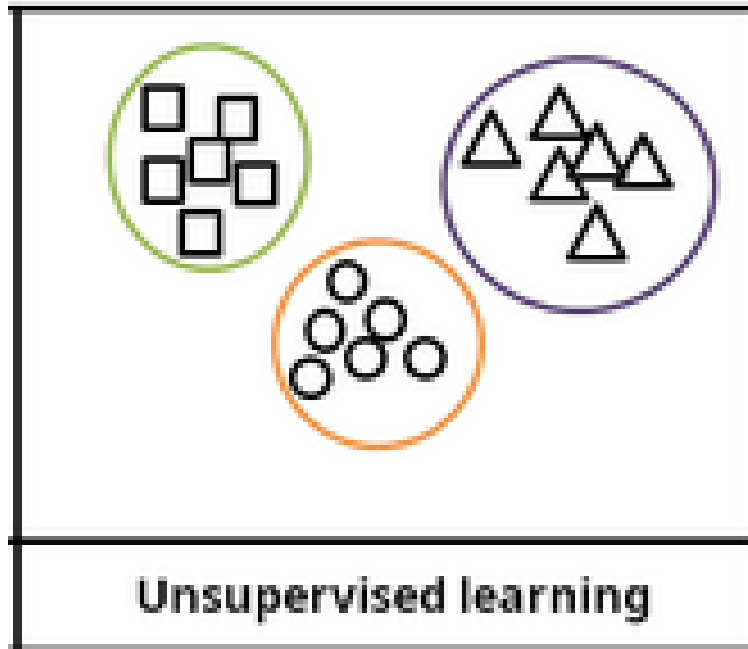
Machine Learning

2. Unsupervised Learning



Machine Learning

2. Unsupervised Learning

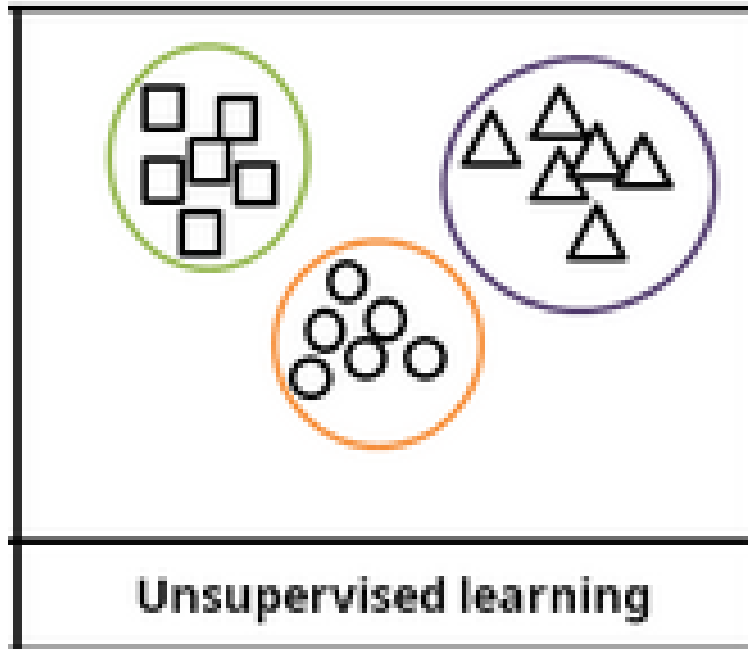


Input data have no accompanying information about the desired output

- Clustering
- Dimensionality reduction

Machine Learning

2. Unsupervised Learning

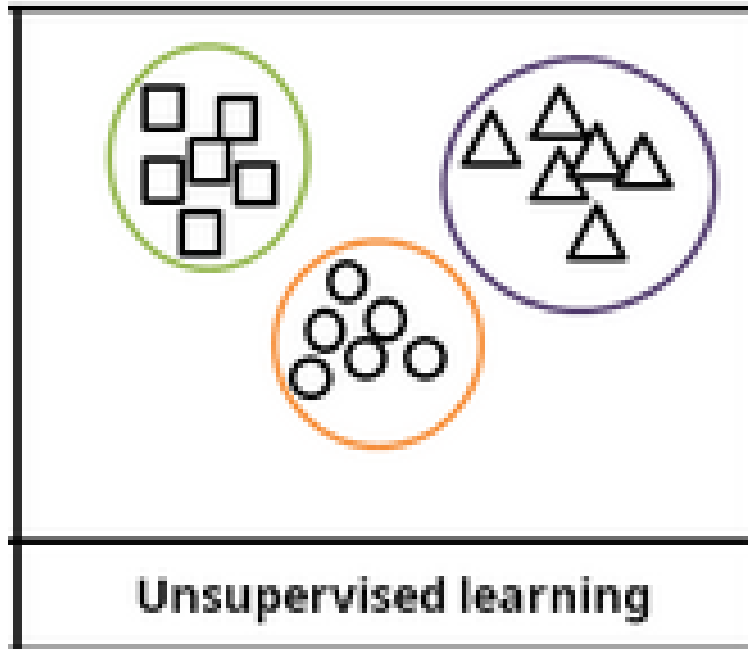


Input data have no accompanying information about the desired output

- **Clustering**
 - k-means,
 - DBSCAN (density-based spatial clustering of applications with noise)
 - Gaussian mixture models

Machine Learning

2. Unsupervised Learning



Input data have no accompanying information about the desired output

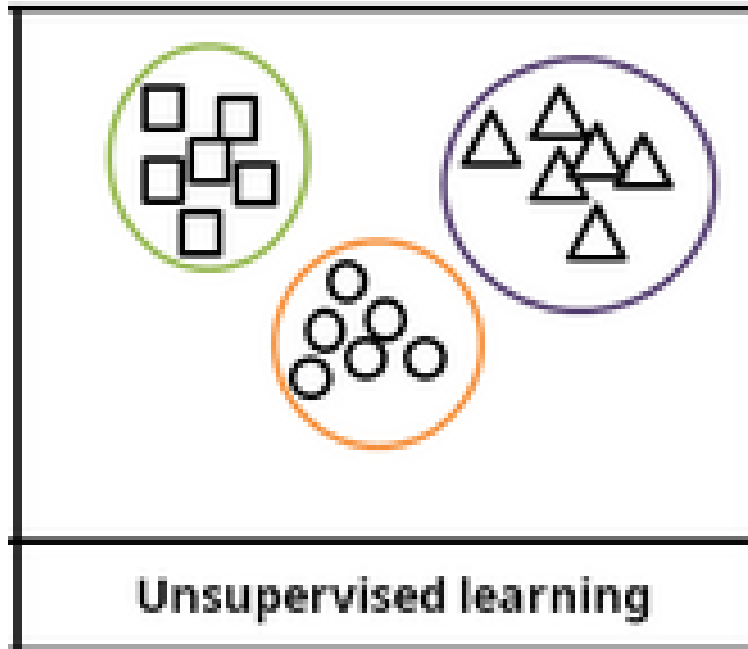
- **Dimensionality reduction**
 - Principal component analysis (PCA)
 - Empirical orthogonal function (EOF)
 - Autoencoder neural networks
 - Contrastive learning models

Machine Learning

2. Unsupervised Learning

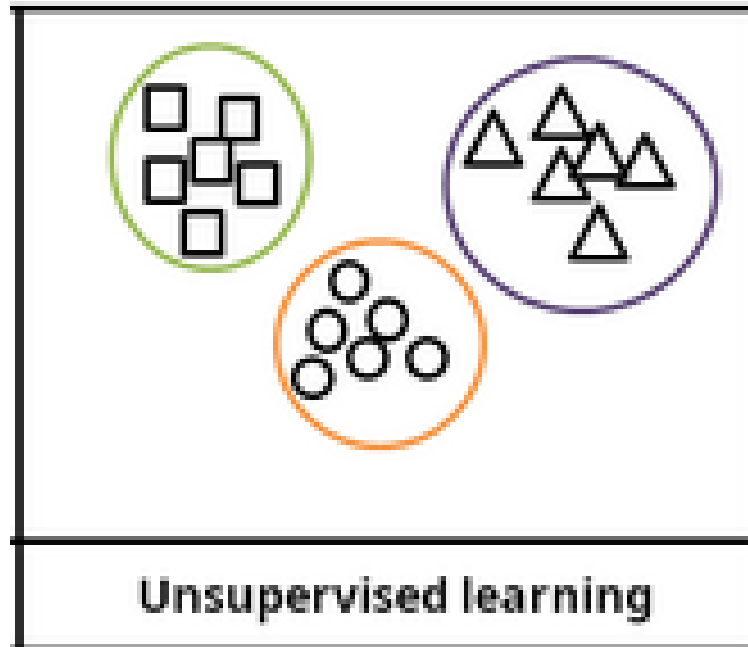
You've done it before: PCA/EOF

- Principal component analysis (PCA)
- Empirical orthogonal function (EOF)
(mathematically the same thing)

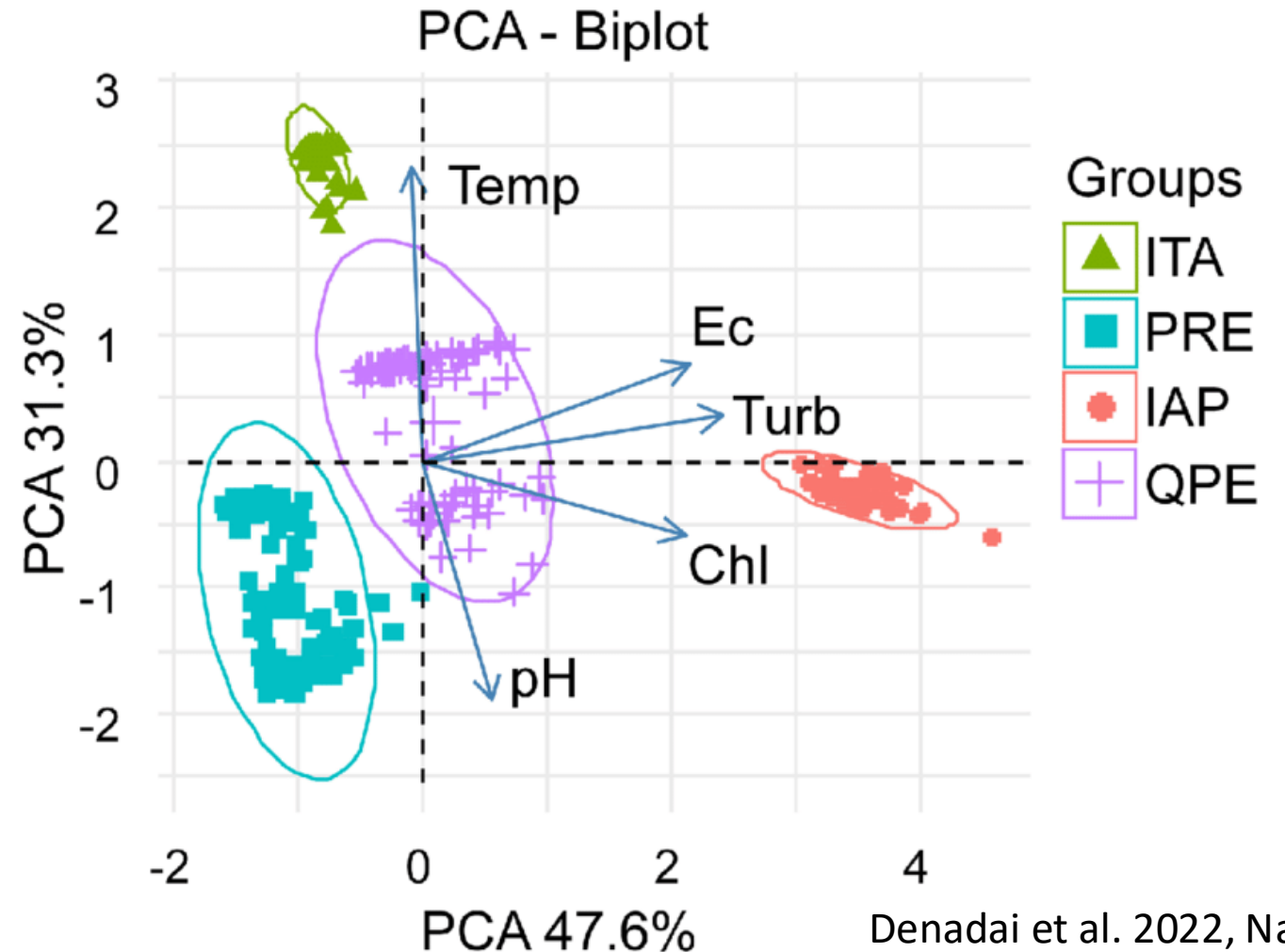


Machine Learning

2. Unsupervised Learning



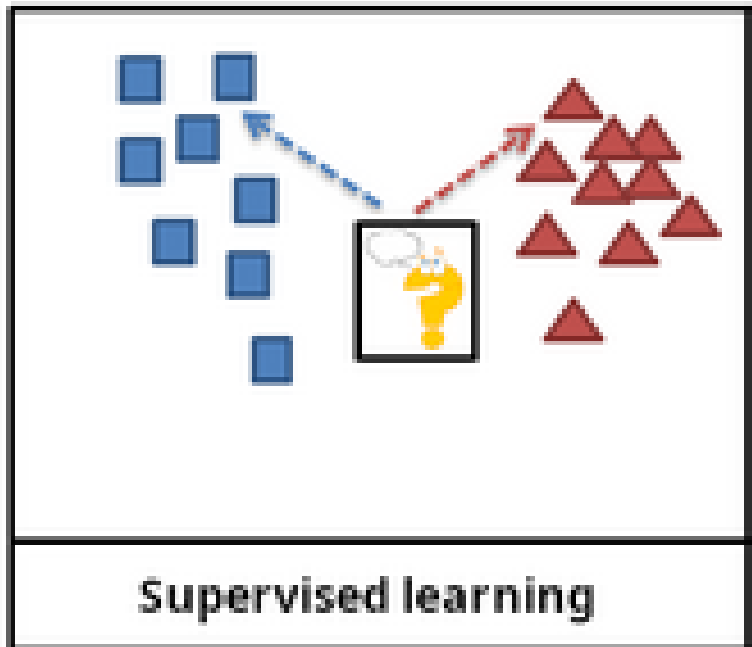
You've done it before: PCA/EOF



Developing Supervised ML Models

1. Supervised Learning

1. Training
2. Validation
3. Testing



Developing Supervised ML Models

1. **Training**
2. Validation
3. Testing

separate labeled subsets train the model

Developing Supervised ML Models

1. Training
- 2. Validation**
3. Testing

model is validated as hyperparameters are tuned (e.g., model size, learning rate, model structure) and the model is re-trained

Developing Supervised ML Models

1. Training
2. Validation
3. **Testing**



**model's performance is tested on
previously unseen data**

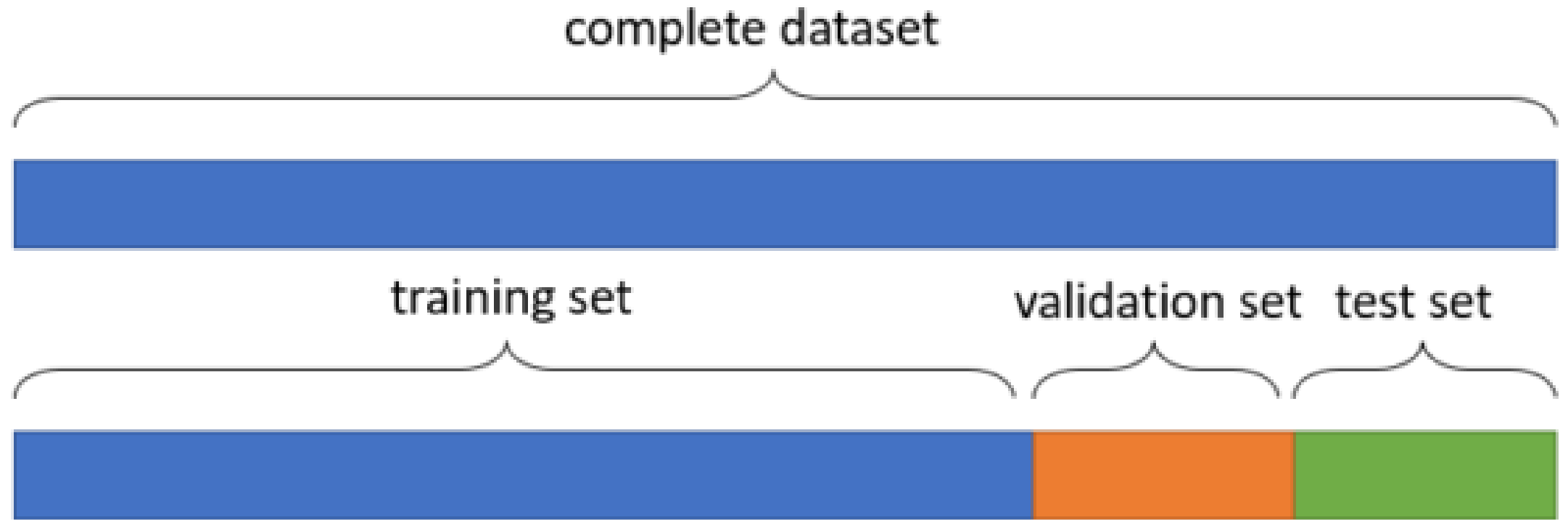
Developing Supervised ML Models

Sometimes Training and Validation are grouped together:

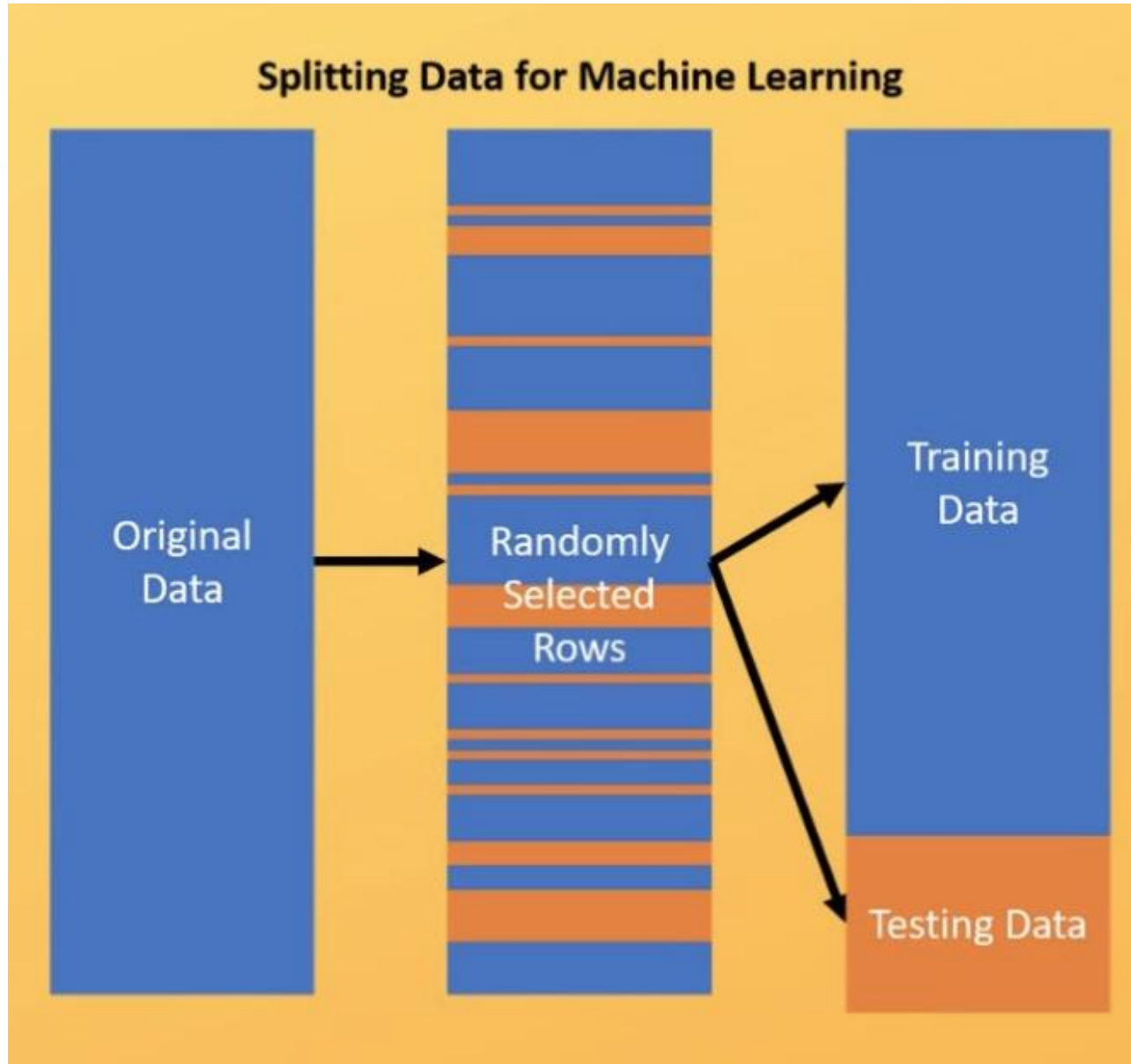
- | | | |
|-------------------|--|-------------------|
| 1. Training |  | 1. Training |
| 2. Validation | | 2. Testing |
| 3. Testing | | |

**model's performance is tested on
previously unseen data**

Developing Supervised ML Models

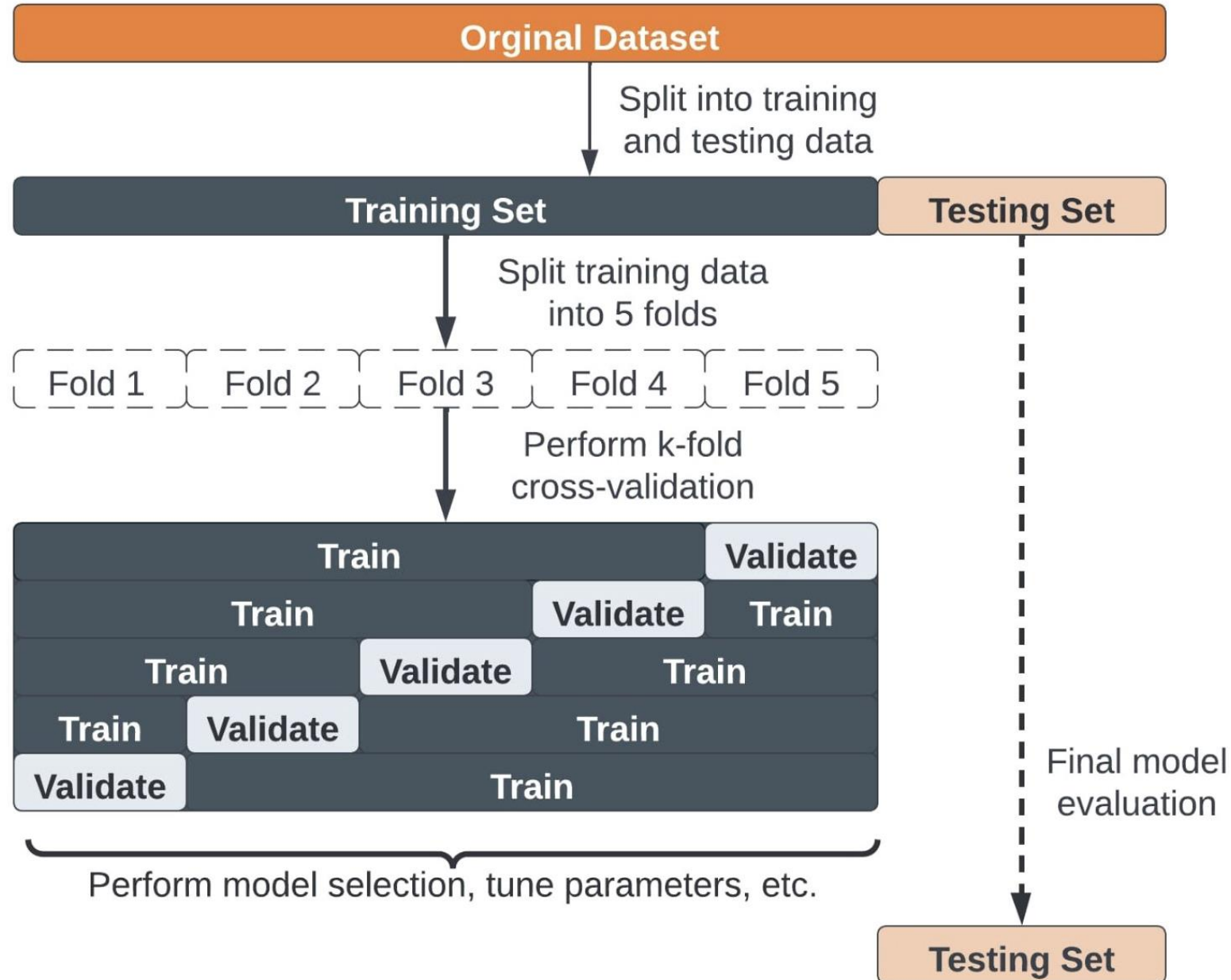


Developing Supervised ML Models



*Set it up so that randomly selected rows are selected the same way every time!

Developing Supervised ML Models



Supervised ML Models for *Regression* that we will try out:

regression_examples.ipynb

- Linear Regressions (ordinary least squares)
- K-Nearest-Neighbors (KNN) Regressions
 - Linear
 - Non-linear
- Multi-Layer Perceptron Regressions
 - Linear
 - Non-linear
- Radius Neighbors Regressions
 - Linear
 - Non-linear